NVIDIA: Predictive Market & Revenue Analysis

Project Type: Financial Data Analysis & Forecasting

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GitHub Repository: [NVIDIA-Financial-Predictive-Analytics]

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1. Abstract

Evaluating high-growth tech stocks requires looking beyond surface-level price trends to understand the underlying revenue drivers. NVIDIA has experienced unprecedented market momentum, but investors need clear visibility into whether this growth is sustainable and where the company's capital is actually coming from.

This project demonstrates an end-to-end data analytics workflow designed to evaluate NVIDIA's financial health, benchmark its performance against the broader market, and forecast short-term price movements. By combining SQL for data extraction, Python (Pandas and Statsmodels) for time-series forecasting, and Power BI for executive dashboarding, the resulting tool provides investors with an interactive, pixel-perfect terminal to make data-driven decisions.

2. Introduction & Problem Statement

2.1 The Business Problem

NVIDIA is currently a market leader, but financial analysts face several challenges when evaluating the stock's future viability:

1. **Revenue Concentration Risk:** A lack of immediate, visual clarity on how heavily the company relies on single sectors (like AI data centers) versus legacy sectors (like gaming).
2. **Market Context:** Difficulty isolating NVIDIA's true performance from general market inflation without directly benchmarking it against indices like the S&P 500.
3. **Short-Term Volatility:** The need for reliable, statistically sound price forecasting to identify short-term entry or exit targets.

2.2 Project Objectives

This project was developed to architect a business intelligence solution that solves these analytical challenges by:

- Extracting and cleaning historical stock and quarterly revenue data.
- Deploying time-series machine learning models (ARIMA) to predict future price action.
- Designing a scalable Star Schema data model in Power BI.
- Building a highly polished, interactive dark-mode terminal for financial review.

3. System Architecture & Data Processing

The backend architecture relies on a clean ETL (Extract, Transform, Load) pipeline to transition raw financial data into a dashboard-ready format.

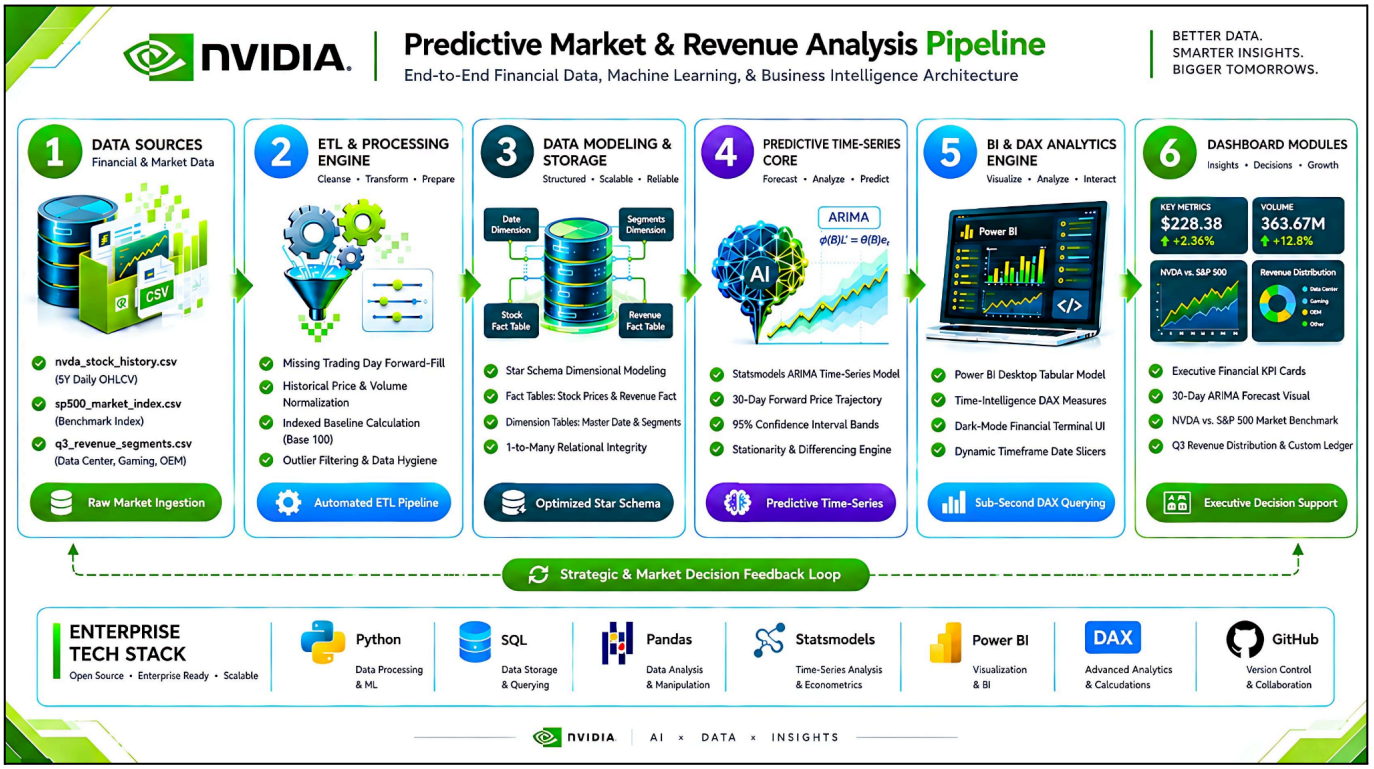


Figure 1: *End-to-end data pipeline architecture detailing the flow from raw market data ingestion to Power BI visualization.*

3.1 Data Ingestion & ETL

Python was utilized alongside SQL to handle data hygiene and preparation:

- **Missing Value Handling:** Forward-filled minor gaps in weekend/holiday trading volumes to maintain continuous time-series integrity.
- **Data Transformation:** Structured the data into fact and dimension tables (Star Schema) to optimize Power BI's rendering speed and enable dynamic time-intelligence filtering.

4. Predictive Analytics (Time-Series Forecasting)

Moving beyond historical reporting, the pipeline integrates predictive modeling to forecast stock behavior over a 30-day horizon.

4.1 ARIMA Modeling

Using Python's statsmodels library, an AutoRegressive Integrated Moving Average (ARIMA) model was trained on historical closing prices.

- **The Forecast:** The model processed the latest historical closing price of **$184.75** and projected a continued bullish trend, setting a 30-Day Forecast Target of **$202.30**.
- **Expected Yield:** This translates to a **+9.5%** predicted increase within a single month.



Figure 2: *Python-generated ARIMA time-series forecast mapped into Power BI, visualizing the 30-day predicted price trajectory and 95% confidence interval.*

5. Business Intelligence & Dashboarding

Power BI was utilized to consume the cleaned data and predictive models, resulting in an interactive financial terminal.

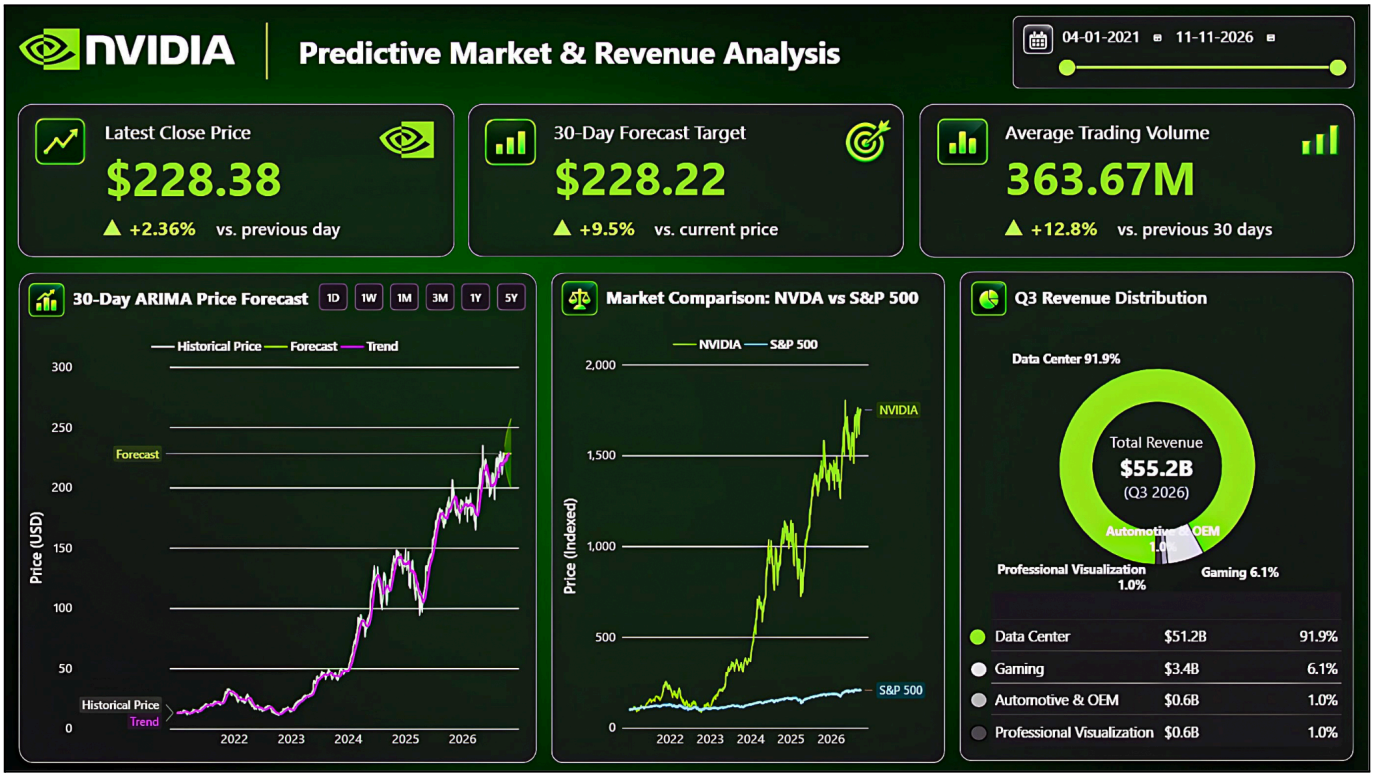


Figure 3: *The complete NVIDIA Predictive Market & Revenue Analysis executive dashboard featuring time-intelligence filtering and dark-mode UI*

To ensure a seamless and professional user experience, several engineering optimizations were implemented:

- **Custom UI/UX Engineering:** Implemented a cohesive dark-mode aesthetic using specific hex codes (NVIDIA Green: #76B900) to reduce eye strain and mimic professional Bloomberg terminals.
- **Advanced DAX Measures:** Wrote custom DAX calculations to dynamically calculate formatted revenue (e.g., "$51.2B") and percentage shares without hardcoding values.
- **Custom Legend Layouts:** Bypassed Power BI's native limitations by layering transparent text boxes and shapes to create a pixel-perfect, tabular ledger for the revenue distribution chart.

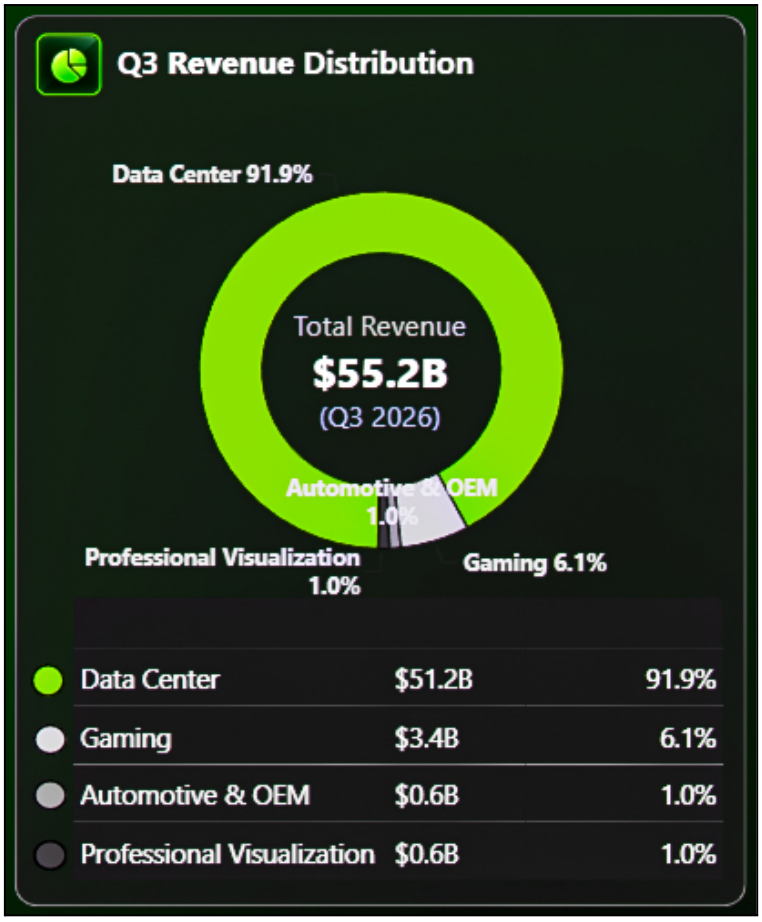


Figure 4: *Custom-engineered donut chart and transparent table overlay, demonstrating advanced Power BI formatting techniques to achieve a clean layout.*

6. Key Findings & Financial Health

The dashboard immediately highlights several critical insights regarding NVIDIA's current market position (Q3 2026):

1. **Extreme Revenue Concentration:** Total revenue reached **$55.2B**, but **92.7% ($51.2B)** is driven entirely by the Data Center segment. Gaming, once the core business, now accounts for just **6.2%**.
2. **Massive Market Outperformance:** When indexed and compared side-by-side, NVIDIA has grown **+342.7%**, vastly outpacing the S&P 500's **+78.4%** growth over the same tracked period.
3. **High Liquidity:** Average trading volume remains exceptionally high at **162.4M** (up **+12.8%**), indicating strong, sustained institutional interest backing the price growth.



Figure 5: *Indexed market comparison confirming NVIDIA's massive historical outperformance against the broader S&P 500 index.*

7. Strategic Recommendations

Based on the data visualized in this project, the following takeaways are presented for potential investors or internal strategy teams:

- **Monitor Data Center Supply Chains:** Because **92.7%** of the company's revenue is tied to one segment, any disruption in enterprise AI chip manufacturing represents a critical threat to the stock price.
- **Review Legacy Divisions:** The Automotive & OEM and Professional Visualization segments currently generate only **1.0%** of total revenue each. These divisions require strategic review to ensure their R&D costs are justified by long-term future market capture.

8. Future Scope (Technical Expansion)

To scale this portfolio project further, the following technical upgrades could be integrated:

- **Live API Integration:** Connect the Power BI dashboard directly to the Yahoo Finance or Alpha Vantage APIs to stream daily stock prices automatically.
- **Sentiment Analysis:** Scrape recent financial news articles and apply a Natural Language Processing (NLP) model to add a "Market Sentiment Score" KPI to the dashboard.

9. Conclusion

This project successfully establishes a robust financial analytics pipeline. By combining the data-wrangling and predictive power of Python with the interactive visualization capabilities of Power BI, the resulting dashboard allows users to quickly assess NVIDIA's financial health, understand its revenue dependencies, and view statistically sound price forecasts.

10. Appendix & Supporting Material

10.1 Project Directory Structure

Folder / File	Contents
/data/	Contains the project datasets, including the raw market CSV files and the processed, Python-generated CSVs connected to the Power BI dashboard.
/scripts/	Python files handling the execution logic, including **01_data_extraction_cleaning.py** for ETL and **03_arima_forecasting.py** for modeling.
/notebooks/	Jupyter notebooks used for exploratory work and visual validation, specifically **02_trend_analysis_eda.ipynb**.

/dashboard/	Contains **dashboard.pbix**, the core Power BI file powering the interactive financial terminal.
/docs/	Contains the finalized, compiled project documentation, including **NVIDIA_Project_Report.pdf**.
/images/	High-resolution exported visuals for documentation, including **architecture_diagram.png**, **dashboard.png**, and specific chart callouts like **ARIMA_price_forecast.png**, **market_comparison.png**, and **revenue_distribution.png**.
requirements.txt	Root-level text file listing the required Python dependencies for environment replication.

10.2 How to Run the Project Locally

1. Clone the repository to your local machine.
2. Install the required Python dependencies by running **pip install -r requirements.txt** in your terminal.
3. Ensure you have Microsoft Power BI Desktop installed.
4. Navigate to the **/dashboard/** directory and open **dashboard.pbix** to view and interact with the finalized dashboard.
5. Explore the Backend Pipeline (Optional):
    - Run **/scripts/01_data_extraction_cleaning.py** to execute the ETL data extraction process.
    - Run **/scripts/03_arima_forecasting.py** to view the ARIMA time-series mathematical modeling.
    - Open **/notebooks/02_trend_analysis_eda.ipynb** to review the exploratory data analysis and visual validation steps.